

DRUM INSTRUMENT Technical Manual

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Purpose: Drum creation, sampling, sequencing, and impact placement. Create and shape drum material, then establish the musical impact and crowd-response field.

Included products: DIH Drum Morpher, BALLPARK, DrumKrushGlue

Routing authority: Drum route: DIH Drum Morpher -> Ballpark -> DrumKrushGlue.

Customer formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64. Standalone applications or other build targets are not included unless a specific installer explicitly lists them.

Canon baseline: 1 = 0.809016994 (PHI/2); standard knob travel is 7 o'clock to 5 o'clock.

DIH Drum Morpher

Sixteen-pad drum sampler and host-synced sequencer with sample loading, pad shaping, restored factory programs, pattern memory, capture, morph, and an original per-pad modeled Layer Engine.

- Product: Drum
- Version: 3.22
- Customer formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64
- Placement: Instrument Track / Drum Source
- Exact DIHEngine call detected: yes
- Output ceiling detected: yes
- Artwork fallback: source/DrMorph.png

OutputNormalizer is prepared before processing and advances rampStep per sample, preventing block-stepped gain changes. The Layer Engine is generated at the host sample rate and enters the same governor, user ceiling, PHI/2 ceiling, sentinel, and optional gain-match stages as the sample path.

State schema 8 stores Engine Mode, Layer, and Variation per pad. Schema 7 sessions remain load-compatible and receive conservative Hybrid defaults when those fields are absent.

Parameter ID	Display name	Type
internal	16 velocity-sensitive pads and MIDI notes	Custom runtime control
internal	WAV, AIFF, and CAF sample loading	Custom runtime control
internal	Volume, pitch, morph, filter, punch, air, and ADSR per pad	Custom runtime control
internal	Engine Mode: Sample / Hybrid / Synth	Per-pad choice
internal	Layer	Per-pad float, 0-1
internal	Variation	Per-pad float, 0-1
internal	Host tempo, swing, grid rate, steps, play, and record	Custom runtime control
internal	Global morph, metamorphosis, momentum bend, governor, and ceiling	Custom runtime control
internal	Sampler capture, playback, and transfer to pad	Custom runtime control
internal	Sixteen per-pad mute toggles for isolating or removing kit voices	Custom runtime control

Meter and visualizer guide

Display	What it measures	How to read it
L / R output	Left and right output activity	Confirms stereo drum output and imbalance.
GOVERNOR	Output containment activity	More activity means protection is holding energy back.
MORPH	Current metamorphosis activity	Shows how strongly the active pads are transforming.
Pad waveform	Loaded sample amplitude over time	Use to identify attacks, tails, and silence in each pad.
Layer readout	Modeled-layer blend for the selected pad	This is reinforcement depth, not an output-gain control.
Variation readout	Deterministic per-hit pitch/tone/pan deviation	Zero is repeat-exact; factory values remain deliberately subtle.

Layer Engine architecture

Pad role	Original model contribution
Kick	Descending membrane oscillator, sub body, and velocity-sensitive beater transient.
Snare	Two shell modes plus a decaying high-frequency wire field.
Clap	Three staggered early noise reflections with a short resonant body.
Rim	Fast dual resonator with a small noise strike.
Closed / open hat	High-passed noise with inharmonic metallic partials and role-specific decay.
Hand percussion	Tuned dual resonators plus velocity-sensitive texture.
Tom / 808	Tuned fundamental, PHI-related upper mode, and short strike noise.
Crash / ride	Long-decay noise field with three inharmonic metallic partials.

Hybrid mode leaves the source sample dominant and adds the model according to Layer. Synth mode lowers the sample contribution and increases the model contribution, but retains enough source imprint to preserve pad identity. Variation is seeded per trigger, so an offline bounce is stable while consecutive hits can differ. Pitch variation is applied to both sample playback and the modeled layer; tone and pan variation remain bounded.

The design was informed by the general modular-drum principle of combining sampled and synthesized components. All oscillator, noise, envelope, preset, and control behavior in this implementation is original DIHWorld code; no third-party sample, patch, or proprietary Reason content is embedded.

BALLPARK

Drum impact and placement processor whose base progression and hit calls turn performance energy into an audible stadium-response field.

- Product: Ballpark
- Version: 3.22
- Customer formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64
- Placement: Drum Track / Drum Bus
- Exact DIHEngine call detected: yes
- Output ceiling detected: yes
- Artwork fallback: Assets/Images/Ballpark_Wide_Stadium.png

Parameter ID	Display name	Type
internal	Base mode and crowd-response progression	Custom runtime control
internal	Mound gain and switch-hitter transient mode	Custom runtime control
internal	Drive, depth, width, focus, clarity, mix, and convergence	Custom runtime control
internal	Input, RMS, peak, gain-reduction, and ceiling meters	Custom runtime control

Parameter ID	Display name	Type
internal	Foul, walk, single, double, triple, and home-run hit calls	Custom runtime control

Meter and visualizer guide

Display	What it measures	How to read it
IN	Input level	Confirms drum energy entering the stadium model.
GR	Gain reduction	Shows active impact containment.
OUT	Output level	Use with IN for level matching.
RMS / PEAK	Body level and fast peak	Their separation describes drum crest and punch.
Ceiling score / hit call	Composite impact, control, and ceiling score	Walk through Home Run describes response intensity, not sports prediction.

DrumKrushGlue

Bus-preferred drum cohesion stage for binding hits into one controlled impact field, with an optional kick-key sidechain that opens a pocket only in its added weight and glue layer.

- Product: DrumKrushGlue
- Version: 3.22
- Customer formats: macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64
- Placement: Drum Bus / Parallel Drum Impact Return
- Exact DIHEngine call detected: yes
- Output ceiling detected: yes
- Artwork fallback: Assets/Images/DrumKrushGlue.png

DrumKrushGlue exposes an optional stereo host Sidechain bus. Its detector is band-limited to approximately 40-180 Hz with a 4 ms attack and user-controlled 40-420 ms recovery. Reduction is applied only to the processed-minus-input Krush/Weight/Glue delta. The main drum path remains outside the duck multiplier. KEY off is sample-path neutral and adds no lookahead or reported latency.

Parameter ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Pocket Duck	Float
sidechain_recovery	Key Recovery	Float

Meter and visualizer guide

Display	What it measures	How to read it
IMPACT	Detected drum peak activity	Shows hit strength entering the impact path.
BODY	Sustained drum energy	Shows weight carried between hits.
REPAIR	Glue/control activity	Shows how strongly cohesion is correcting the bus.
CEILING	Protected output pressure	Persistent maximum activity means the chain is overdriven.
KEY	Detected 40-180 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
POCKET	Reduction of the added weight/glue layer	The original drum path remains outside this reduction.